Numeric data types

DATA TYPES IN PYTHON

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Built in numeric types

Integer

- Whole numbers
- Large values

```
int(1234567891234567 89)
```

```
1234567891234567 89
```

Float

- Fractional amounts (approximation)
- Scientific notation

```
float(1234567891234567 89)
```

```
1.2345678912345678e+17
```

Decimals

- Exact precision
- Currency operations

```
from decimal import Decimal
Decimal('1234567891234567 89')
```

```
Decimal('1234567891234567 89')
```

Printing floats

```
print(0.00001)
```

```
1e-05
```

```
print(f"{0.00001:f}")
```

```
0.000010
```

Printing floats

```
print(f"{0.0000001:f}")
```

```
0.000000
```

```
print(f"{0.0000001:.7f}")
```

```
0.0000001
```

Python division types

```
4/2
```

```
2.0
```

```
4//2
```

```
2
```

```
7//3
```

```
2
```

Let's practice!

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Booleans - the logical data type

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Booleans as a data type

- `True`
- `False`

Notice the capitalization as this can trip you up when switching between Python and other languages.

```
out_of_cookies = True
if out_of_cookies:
    print("Run to the store NOW!")
```

```
Run to the store NOW!
```

Truthy and Falsey

- Truthy values are ones that will return true
- Falsey values will evaluate to false

```
apples=2
if apples:
    print("We have apples.")
```

```
"We have apples."
```

```
apples=0
if apple:
    print('We have apples.')
```

Truthy and Falsey

Truthy

- `1`
- `"Cookies"`
- `["Cake", "Pie"]`
- `{"key": "value"}`

Falsey

- `0`
- `""`
- `[]`
- `{}`
- `None`

Operators - a boolean evaluation context

```
cookie_qty == 3
```

- `==` equal to
- `!=` not equal to
- `<` less than
- `<=` less than or equal to
- `>` greater than
- `>=` greater than or equal to

Floats are approximately an issue

```
x = 0.1 + 1.1
x == 1.2
```

```
False
```

```
print(x)
```

```
1.2000000000000002
```

Be careful with equality comparisons of floats!

Let's practice!

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Sets (unordered data with optimized logic operations)

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Set

- Unique
- Unordered
- Mutable
- Python's implementation of Set Theory from Mathematics

Creating sets

- Sets are created from a list

```
cookies_eaten_today = ['chocolate chip', 'peanut butter',
   ...: 'chocolate chip', 'oatmeal cream', 'chocolate chip']
types_of_cookies_eaten = set(cookies_eaten_today)
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'oatmeal cream', 'peanut butter'])
```

Modifying sets

- `.add()` adds single elements

```
types_of_cookies_eaten.add('biscotti')

types_of_cookies_eaten.add('chocolate chip')

print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'oatmeal cream', 'peanut butter', 'biscotti'])
```

Updating sets

- `.update()` merges in another set or list

```
cookies_hugo_ate = ['chocolate chip', 'anzac']

types_of_cookies_eaten.update(cookies_hugo_ate)

print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream',  'peanut
```

Removing data from sets

- `.discard()` safely removes an element from the set by value
- `.pop()` removes and returns an arbitrary element from the set (KeyError when empty)

```
types_of_cookies_eaten.discard('biscotti')
print(types_of_cookies_eaten)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream', 'peanut butter'])
```

```
types_of_cookies_eaten.pop()
types_of_cookies_eaten.pop()
```

```
'chocolate chip'
'anzac'
```

Set operations - similarities

- `.union()` set method returns a set of all the names (`or`)
- `.intersection()` method identifies overlapping data (`and`)

```
cookies_jason_ate = set(['chocolate chip', 'oatmeal cream',
'peanut butter'])
cookies_hugo_ate = set(['chocolate chip', 'anzac'])
cookies_jason_ate.union(cookies_hugo_ate)
```

```
set(['chocolate chip', 'anzac', 'oatmeal cream', 'peanut butter'])
```

```
cookies_jason_ate.intersection(cookies_hugo_ate)
```

```
set(['chocolate chip'])
```

Set operations - differences

- `.difference()` method identifies data present in the set on which the method was used that is not in the arguments (`-`)
- Target is important!

```
cookies_jason_ate.difference(cookies_hugo_ate)
```

```
set(['oatmeal cream', 'peanut butter'])
```

```
cookies_hugo_ate.difference(cookies_jason_ate)
```

```
set(['anzac'])
```

Let's practice!

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